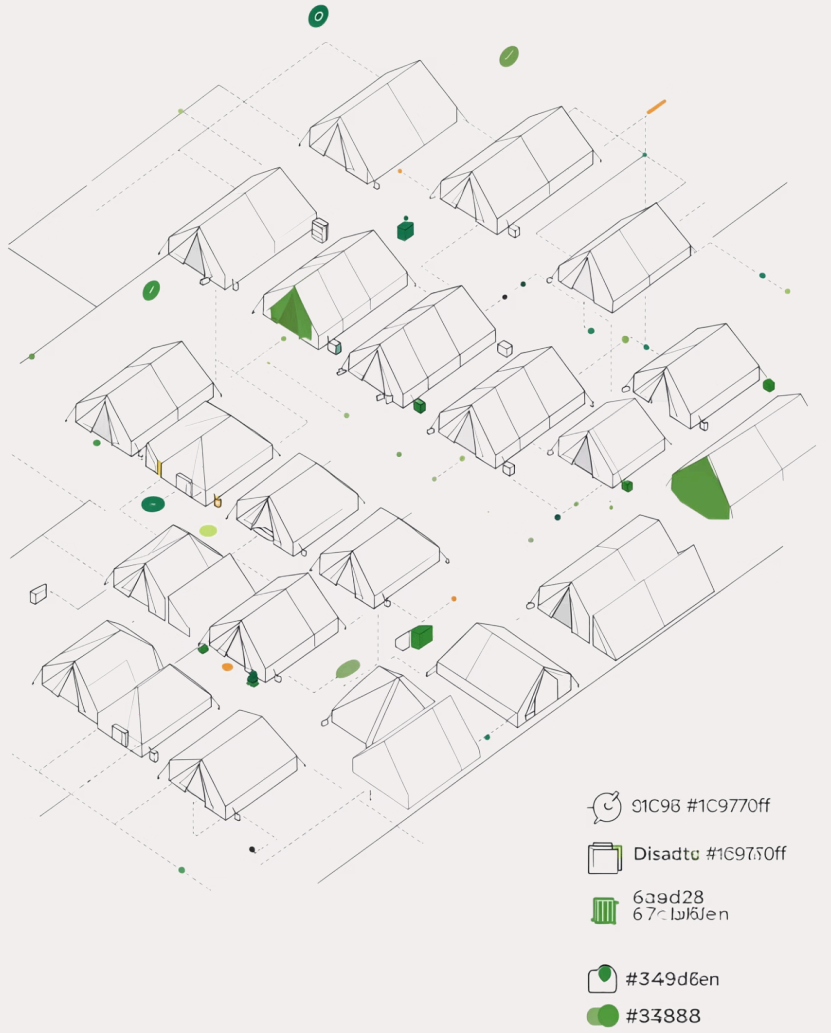
Disaster Relief Resource & Camp Management System

Database Design & Implementation Project — Review 1



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The Problem: Manual Coordination Failures

Fragmented Records

Camps, families, needs, stock, donations, and volunteers managed through disconnected manual lists — no unified data source.

Duplicate Entitlements

Families receive supplies more than once while others go without — unequal distribution with no detection mechanism.

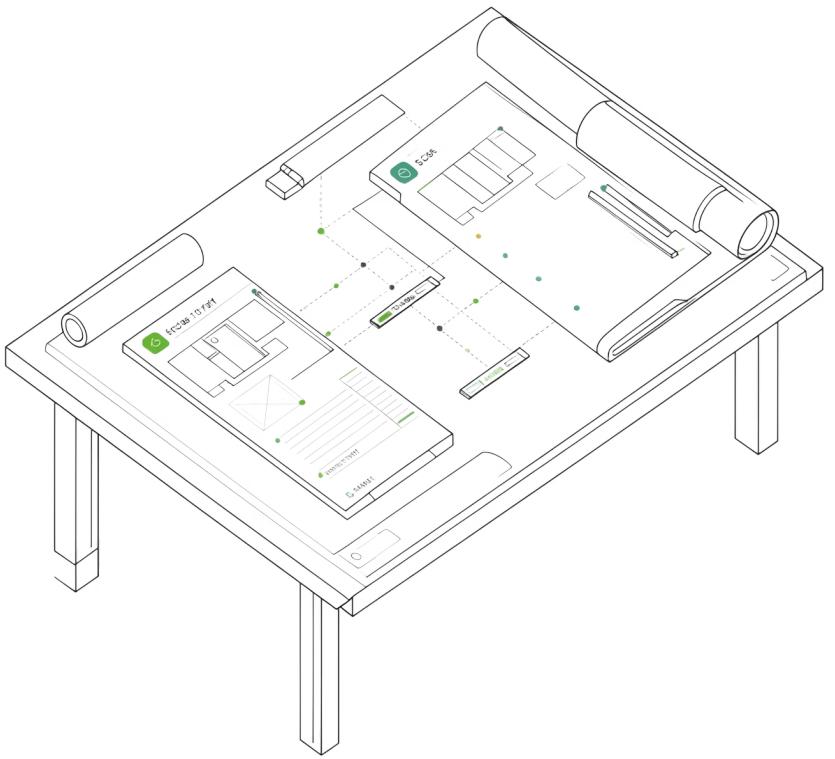
No Traceability

Stock movement and volunteer assignments are untracked, making auditing and accountability impossible.

Inefficient Allocation

Resources cannot be matched to needs across multiple locations, leading to waste and shortages simultaneously.

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Project Scope & Objectives

1 Scope

Design a relational database covering disaster relief operations across **multiple camps and locations**.

2 Key Objectives

Eliminate duplication, ensure equitable distribution, enable full resource and volunteer tracking.

3 Deliverable

Normalized **3NF schema** with integrity constraints, representative sample data, and SQL queries.

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Core Entities & Relationships

Fourteen principal entities organized across four operational categories.



Operational

- Disasters
- Locations
- Camps
- Warehouses



People

- Families
- Persons
- Volunteers



Resources

- Relief Items
- Stock
- Donations

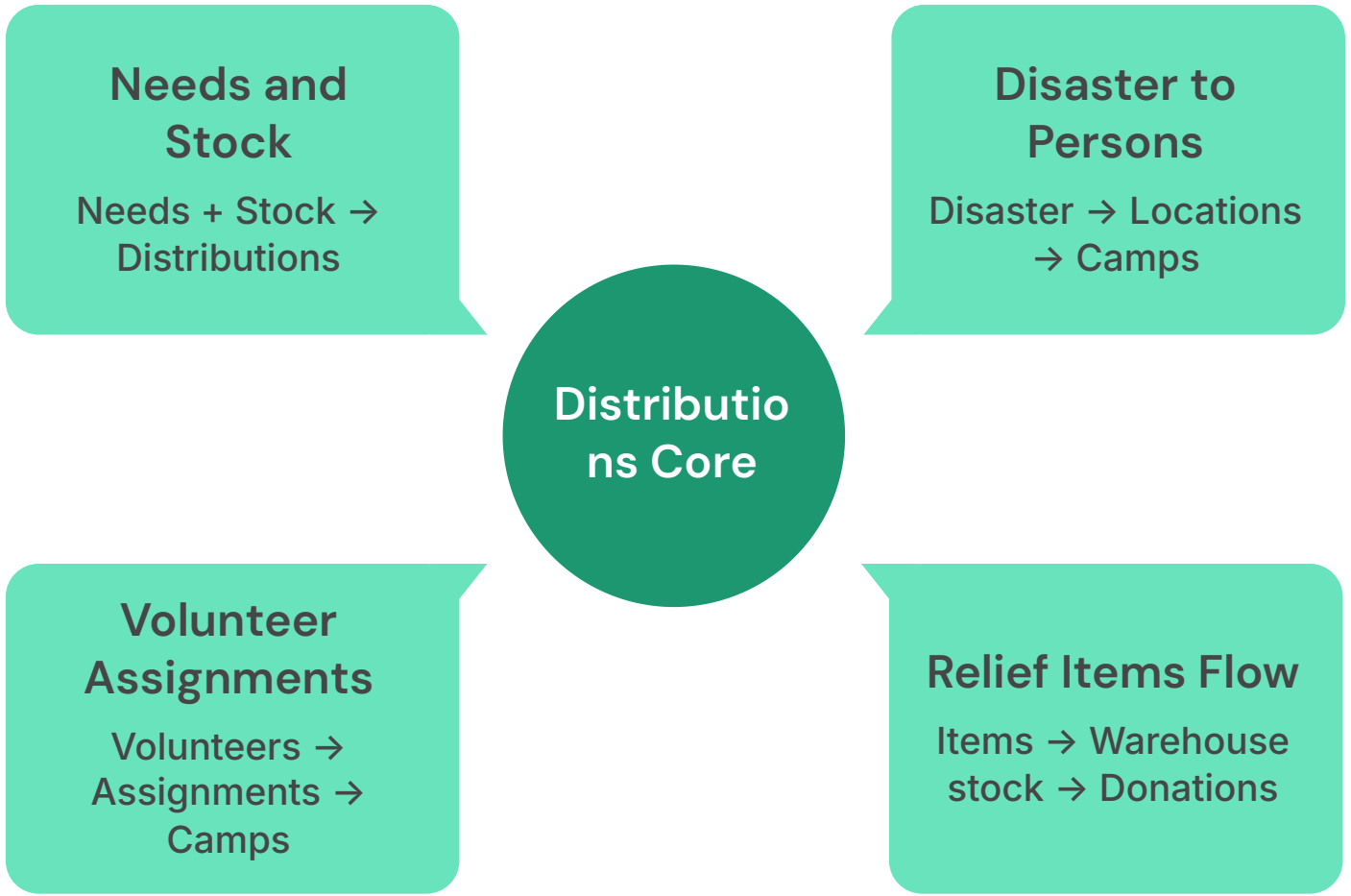


Activities

- Needs
- Distributions
- Assignments
- Referrals

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ER Model: Key Relationships



Each chain reflects a distinct operational domain — all four converge at the **Distributions** entity, the system's core transactional record.

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Relational Schema: Critical Tables

Table	Primary / Foreign Keys	Key Attributes
Families	FamilyID *(PK)*, CampID *(FK)*	HeadOfFamily, ContactInfo
Needs	NeedID *(PK)*, FamilyID *(FK)*, ItemID *(FK)*	Quantity, Status
Stock	StockID *(PK)*, ItemID *(FK)*, WarehouseID *(FK)*	Quantity, LastUpdated
Distributions	DistributionID *(PK)*, NeedID *(FK)*, StockID *(FK)*	Quantity, Date

All tables normalized to **3NF** — no transitive dependencies, all non-key attributes depend solely on the primary key.

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